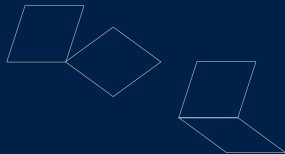


Nakayama Algebras

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MTH619 2025-26 I - Endsemester Presentation

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- ▶ Composition series and composition length
- ▶ Radical series, socle series, radical length, socle length and Loewy length
- ▶ Uniserial modules, right serial and left serial algebras
- ▶ Ordinary quiver of an algebra
- ▶ Nakayama Algebras
 1. Indecomposables in Nakayama Algebras
 2. Almost Split Sequences for Nakayama Algebras



Definition 1

Given a finite dimensional K -algebra, A , and M in $\text{mod-}A$, the chain

$$0 = M_0 \subset M_1 \subset M_2 \subset \dots \subset M_m = M$$

of submodules of M such that M_{j+1}/M_j is simple for $j = 0, 1, \dots, m-1$ is called a **composition series** of M .

- Every such module M has a composition series.



Composition Series

Theorem 2

Jordan - Hölder Theorem

Given two composition series for a module M ,

$$0 = M_0 \subset M_1 \subset M_2 \subset \dots \subset M_m = M,$$

$$0 = N_0 \subset N_1 \subset N_2 \subset \dots \subset N_n = M$$

$m = n$ and there is a permutation σ of $\{1, \dots, m\}$ such that $M_{j+1}/M_j \cong N_{\sigma(j+1)}/N_{\sigma(j)}$ for any $j = 0, 1, \dots, m - 1$.

- ▶ Hence, the number of modules in the composition series depends only on M . We call this the **composition length** of M , and denote it by $l(M)$.
- ▶ The composition length, $l(M)$, is also just the sum of the elements in $\dim M$.



Definition 3

For a right A -module, M , the (Jacobson) **radical** is the intersection of all the maximal submodules of M , and is denoted by $\text{rad}M$.

Definition 4

For a right A -module, M the **socle** is the submodule of M that is generated by all simple submodules of M , and is denoted by $\text{soc}M$.

Definition 5

For a right A -module, M , $\text{top}M = M/\text{rad}M$, is called the **top** of M .



Descending Loewy Series of M

Definition 6

For an A -module, M , the decreasing sequence of submodules of M ,

$$M \supset \text{rad}M \supset \text{rad}^2M \supset \dots \supset \text{rad}^iM \supset \dots \supset 0$$

is called the **radical series** or **descending Loewy series** of M .

- ▶ Since M is finite dimensional, the series is finite.
- ▶ Let m be the least positive integer such that $\text{rad}^mM = 0$. This m is called the **length of the radical series**, and is denoted by $rl(M)$.



Ascending Loewy Series of M

For an integer $i \geq 0$, we define $\text{soc}^i(M)$ inductively, with $\text{soc}^0 M = \text{soc} M$. Let p be the quotient map, $p : M \rightarrow M/\text{soc}^i M$. Then $\text{soc}^{i+1} M = p^{-1}(\text{soc}(M/\text{soc}^i M))$. Note that $\text{soc}^{i+1} M \supset \text{soc}^i M$.

Definition 7

For an A -module, M , the increasing sequence of submodules of M ,

$$0 = \text{soc}^0 M \subset \text{soc} M = \text{soc}^1 M \subset \text{soc}^2 M \subset \dots \subset \text{soc}^i M \subset \dots \subset M$$

is called the **socle series** or **ascending Loewy series** of M .

- ▶ Since M has finite composition length, the series is finite.
- ▶ Let m be the least positive integer such that $\text{soc}^m M = M$. This m is called the **length of the socle series**, and is denoted by $sl(M)$.



Relation between $rl(M)$ and $sl(M)$ and Loewy Length

Radical Series:

$$M \supset \text{rad}M \supset \text{rad}^2M \supset \dots \supset \text{rad}^iM \supset \dots \supset 0$$

Socle Series:

$$0 = \text{soc}^0M \subset \text{soc}M = \text{soc}^1M \subset \text{soc}^2M \subset \dots \subset \text{soc}^iM \subset \dots \subset M$$

- ▶ In general, the radical series and the socle series don't coincide.
- ▶ However, $rl(M) = sl(M)$!
- ▶ We call this common value of $rl(M)$ and $sl(M)$ as the **Loewy length** of M , denoted by $ll(M)$.



Some useful results

- ▶ $M\text{rad}A = \text{rad}M$
- ▶ $\text{rad}(M \oplus N) = \text{rad}M \oplus \text{rad}N$
- ▶ Given an A -module epimorphism $f : M_A \rightarrow N_A$, $f(\text{rad}^i M) = \text{rad}^i N$ for every $i \geq 0$.
- ▶ Given a decomposition of a module $M = M_1 \oplus \dots \oplus M_m$, we have

$$\text{ll}(M) = \max\{\text{ll}(M_1), \dots, \text{ll}(M_m)\}$$



Definition 8

An A -module M_A is called **uniserial** if it has a unique composition series.

- ▶ If M is uniserial, every submodule of M is also uniserial.
- ▶ If M is uniserial, every quotient of M is also uniserial.
- ▶ Uniserial modules are determined up to isomorphism by their composition series.
- ▶ The dual of a uniserial right module is a uniserial left module.
- ▶ Uniserial modules have simple tops, and simple socles.
- ▶ All uniserial modules are indecomposable.



Lemma 9

The following are equivalent for M , a right A -module:

1. M is uniserial.
2. The radical series, $M \supset \text{rad}M \supset \dots \supset 0$ is a composition series.
3. The socle series, $0 \subset \text{soc}M \subset \dots \subset M$ is a composition series.
4. $l(m) = ll(m)$



Radical, Socle and composition series for uniserial modules

Proof.

(1 $\implies$ 2)

Induction on composition length $l(M)$ of M :

- Induction Step: Consider a uniserial module M , with $l(M) = t$. Since M is uniserial, it has a unique maximal submodule, coinciding with $\text{rad}M$, which is also uniserial. The result now follows from the induction hypothesis.

(2 $\implies$ 1)

Assume $M = M_0 \supset M_1 \supset \dots \supset M_t = 0$ and $M = N_0 \supset N_1 \supset \dots \supset N_t = 0$

are two composition series for M . Then, $M_i = N_i = \text{rad}^i M$ for every $0 \leq i \leq t$.

- Induction Step: $\text{rad}^i M / \text{rad}^{i+1} M$ is simple $\implies \text{rad}^i M$ has a unique maximal submodule, which is necessarily $\text{rad}^{i+1} M \implies M_{i+1} = N_{i+1} = \text{rad}^{i+1} M$.

(1 $\iff$ 3) is analogous.



Proof.

(4 $\implies$ 2)

- ▶ If N is an A -submodule of M , then $l(M) = l(N) + l(M/N)$ (Follows from the Grothendieck Group structure - consider the short exact sequence:
 $0 \rightarrow N \rightarrow M \rightarrow M/N \rightarrow 0$).
- ▶ Hence, $m = l(M) = ll(M) = \sum_{i=0}^{m-1} l(\text{rad}^i M / \text{rad}^{i+1} M)$
 $\implies l(\text{rad}^i M / \text{rad}^{i+1} M) = 1$ for $i = 0, 1, \dots, m-1$ (since they can't be 0).
- ▶ This implies that each $\text{rad}^i M / \text{rad}^{i+1} M$ is simple, implying that the radical series is a composition series.

(2 $\implies$ 4) is obvious.





Definition 10

An algebra A is **right serial** if every indecomposable projective right A -module is uniserial.

Definition 11

An algebra A is **left serial** if every indecomposable projective left A -module is uniserial.

Equivalently,

- ▶ A is right serial if every indecomposable injective left A -module is uniserial.
- ▶ A is left serial if every indecomposable injective right A -module is uniserial.



Lemma 12

An algebra A , is right serial if and only if for every indecomposable projective right module P , the module $\text{rad}P/\text{rad}^2P$ is simple or zero.

Idea of Proof.

($\implies$)

$$M \supset \text{rad}P \supset \text{rad}^2P \supset \dots \supset \text{rad}^iP \supset \dots \supset 0$$

The radical series of uniserial modules is a composition series.

($\impliedby$)

We can show that the radical series is a composition series by proving that $\text{rad}^{i-1}P/\text{rad}^iP$ is simple or zero for $i \geq 1$, using induction on i .



Ordinary Quiver of an Algebra

Let A be a basic and connected finite dimensional K -algebra and $\{e_1, e_2, \dots, e_n\}$ be a complete set of primitive orthogonal idempotents of A . The **ordinary quiver** of A , Q_A is defined as follows:

1. The n vertices of the quiver correspond to $e_1, e_2, \dots, e_n$.
2. Given $a, b \in (Q_A)_0$, the arrows $\alpha : a \rightarrow b$ correspond to the vectors in a basis of $e_a(\text{rad}A/\text{rad}^2A)e_b$. The dimension of this vector space is the number of arrows going from a to b .

An algebra is called **basic**, if $e_i A \not\cong e_j A$ for all $i \neq j$.

An algebra is called **connected** if the only central idempotents are 0 and 1.



Ordinary Quiver of an Algebra

For example,

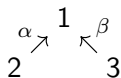
$$\text{Consider the algebra } A = \begin{bmatrix} K & 0 & 0 \\ K & K & 0 \\ K & 0 & K \end{bmatrix}$$

of lower triangular matrices $\lambda_{ij} \in M_3(K)$, with $\lambda_{32} = 0$ and $\lambda_{pq} = 0$ for $p > q$.

A complete set of primitive orthogonal idempotents is:

$$e_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, e_2 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}, e_3 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}. \text{ Also, } \text{rad}A = \begin{bmatrix} 0 & 0 & 0 \\ K & 0 & 0 \\ K & 0 & 0 \end{bmatrix}$$

and $\text{rad}^2 A = 0$. $e_2(\text{rad}A)e_1$ and $e_3(\text{rad}A)e_1$ are one-dimensional, and all other spaces of the form $e_i(\text{rad}A)e_j$ are 0, giving Q_A as,





Characterization of Basic Right Serial Algebras

Theorem 13

A basic K -algebra is right serial if and only if, for every point a of its ordinary quiver Q_A , there exists at most one arrow of source a .

Proof.

- ▶ Follows from Lemma 12.
- ▶ $\text{rad}P(a)/\text{rad}^2P(a) = e_a(\text{rad}A/\text{rad}^2A)$ is simple or zero, and hence as a K -vector space has a dimension of 0 or 1.
- ▶ By definition of Q_A , there is at most one $b \in (Q_A)_0$ such that $e_a(\text{rad}A/\text{rad}^2A)e_b \neq 0$, i.e. there is at most one such b with an arrow $a \rightarrow b$.





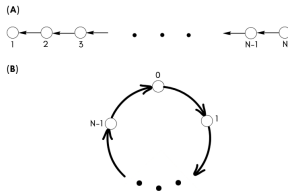
Nakayama Algebras

Definition 14

An algebra, A , is called a **Nakayama Algebra** if it is both right and left serial.

Theorem 15

A basic and connected algebra A is a Nakayama Algebra if and only if its ordinary quiver Q_A is one of the following:



Proof follows from the previous characterization.



Indecomposables in Nakayama Algebras

Lemma 16

Let A be an algebra, and J , a proper ideal of A .

- 1. If A is right serial, then so is A/J .*
- 2. If A is a Nakayama Algebra, then so is A/J .*

Idea of Proof.

- ▶ (1) follows from the direct sum decomposition of indecomposables.
- ▶ (2) follows from (1) and it's dual.



Indecomposables in Nakayama Algebras

Theorem 17

Let A be a basic and connected Nakayama algebra. For any indecomposable A -module M , there is an indecomposable projective P and an integer t with $1 \leq t \leq \text{ll}(P)$ such that $M \cong P/\text{rad}^t P$.

Idea of Proof.

- ▶ Take $t = \text{ll}(M)$. This gives $\text{rad}^t M = 0 \implies M \text{rad}^t A = 0$. So, M has the structure of an $A/\text{rad}^t A$ module, which is itself a Nakayama algebra, with the decomposition, $A/\text{rad}^t A \cong \bigoplus_{i=1}^n (P_i/P_i \text{rad}^t A) = \bigoplus_{i=1}^n (P_i/\text{rad}^t P_i)$
- ▶ Let $f : \bigoplus_{j=1}^r P'_j \rightarrow M$ be a projective cover of M in $A/\text{rad}^t A$, with P'_j indecomposable for all $j = 1, 2, \dots, r$.
- ▶ $\exists q \in \{1, 2, \dots, r\}$ such that $\text{ll}(P'_q) = t$ and $f_q = f|_{P'_q}$ is a monomorphism, which then gives us $M \cong P'_q \cong P_i/\text{rad}^t P_i$ for some $1 \leq i \leq n$

★ As a consequence, **all indecomposables are uniserial!**



Almost Split Sequences for Nakayama Algebras

Theorem 18

For an indecomposable nonprojective A -module, $M \cong P/\text{rad}^t P$, the following is an almost split sequence.

$$0 \rightarrow \text{rad} P / \text{rad}^{t+1} P \xrightarrow{[q,i]^t} (\text{rad} P / \text{rad}^t P) \oplus (P / \text{rad}^{t+1} P) \xrightarrow{[-j,p]} P / \text{rad}^t P \rightarrow 0$$

where q, p are the canonical epimorphisms (quotient maps), and i, j are inclusion homomorphisms.

Corollary 19

For every indecomposable nonprojective A -module M , we have $l(\tau M) = l(M)$.

★ All almost split sequences in the module category of Nakayama Algebras have at most two indecomposable middle terms.



Where are Nakayama Algebras used?

- ▶ They are used to study regular modules over representation-infinite hereditary algebras.
- ▶ Derived categories of Nakayama Algebras are used in **algebraic geometry**.
- ▶ Linear Nakayama Algebras correspond to the **combinatorics** of Dyck paths via their AR Quiver.



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